

AWS CLOUD SECURITY & SECURITY MONITORING LAB



S3 Security, CloudTrail Data-Event Monitoring, CloudWatch Detection and SNS Alerting

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1. Introduction

Cloud environments provide scalable infrastructure and services, but they also introduce security challenges related to access control, data protection, monitoring, and the detection of suspicious activity.

This project implements a practical AWS cloud security monitoring environment using Amazon S3, AWS CloudTrail, Amazon CloudWatch, and Amazon SNS. The lab focuses on monitoring object-level activity within an S3 bucket and generating an automated security alert when an object is uploaded.

The implemented workflow demonstrates how a cloud security event can be collected, analysed, and converted into an actionable alert. An S3 `PutObject` operation is recorded as a CloudTrail data event, forwarded to CloudWatch Logs, identified using a CloudWatch metric filter, and used to trigger a CloudWatch alarm. Amazon SNS then delivers an email notification when the defined security condition is detected.

The project provides practical experience with AWS security controls, cloud audit logging, security monitoring, event detection, and automated alerting.

2. Objectives

The main objectives of this project are to:

- Configure a secure Amazon S3 bucket using appropriate security controls.
- Enable AWS CloudTrail data-event logging to monitor object-level activity within the S3 bucket.
- Integrate CloudTrail with Amazon CloudWatch Logs for centralized security monitoring.
- Create a CloudWatch metric filter to detect S3 `PutObject` events.
- Configure a CloudWatch alarm to identify detected upload activity.
- Use Amazon SNS to automatically send an email security alert when the alarm is triggered.
- Demonstrate an end-to-end cloud security monitoring and alerting workflow.

3. AWS Environment

The lab was implemented using the following AWS services and components:

Component	Purpose
Amazon S3	Secure storage and generation of object-level activity
AWS CloudTrail	Records S3 data events such as PutObject
Amazon CloudWatch Logs	Centralizes CloudTrail security events
CloudWatch Metric Filter	Detects PutObject events
CloudWatch Alarm	Triggers when a detected event meets the defined threshold
Amazon SNS	Sends an automated email security notification

AWS Region

The primary S3 lab bucket was deployed in Europe (Stockholm) — **eu-north-1** AWS Region.

Lab Architecture

The overall security monitoring workflow implemented in this project is:

S3 Object Upload → CloudTrail → CloudWatch Logs → Metric Filter → CloudWatch Metric → CloudWatch Alarm → SNS Email Alert

4. Secure S3 Bucket Configuration

An Amazon S3 bucket was configured as the primary storage resource for the security monitoring lab. Security-focused configuration settings were applied to reduce the risk of

unauthorized public access and to provide protection and accountability for stored objects.

4.1 Bucket Versioning

Versioning was enabled on the S3 bucket. This allows multiple versions of an object to be retained when objects are modified or replaced.

Bucket overview

AWS Region
Europe (Stockholm) eu-north-1

Amazon Resource Name (ARN)
 arn:aws:s3:::security-lab-fawwaz-2026-569486438195-eu-north-1-an

Creation date
October 3, 2026, 14:17:37 (UTC+03:00)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)
Disabled

Default encryption

Edit

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)
Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key
When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)
Enabled

Blocked encryption types [Info](#)
Server-Side Encryption with Customer-Provided Keys (SSE-C)

Figure 1 — S3 Bucket Properties: Versioning and Encryption

4.2 Server-Side Encryption

The bucket uses Amazon S3 managed encryption keys (SSE-S3) for server-side encryption. S3 Bucket Key is also enabled.

This provides encryption for objects stored within the bucket without requiring application-level encryption.

4.3 Block Public Access

All S3 Block Public Access settings were enabled. This prevents the bucket and its objects from being unintentionally exposed through public access configurations.

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for eu-north-1](#)

Block public access (bucket settings)

[Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

On

Individual Block Public Access settings for this bucket

☒ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Figure 2 : S3 bucket permissions showing Block Public Access enabled and bucket ownership controls.

4.4 Object Ownership

Bucket owner enforced object ownership was configured, with ACLs disabled. This ensures that access control is managed through IAM and bucket policies rather than traditional S3 object ACLs.

Security Significance

The S3 configuration establishes a secure baseline for the lab by combining:

- Object versioning
- Server-side encryption
- S3 Bucket Key
- Block Public Access
- Bucket owner enforced object ownership

These controls provide foundational data protection, access control, and secure storage before monitoring and detection mechanisms are introduced.

5. AWS CloudTrail Data Event Logging

AWS CloudTrail was configured to provide object-level monitoring for activity within the S3 lab bucket. Unlike management events, S3 data events can record operations performed on individual objects, such as uploading, retrieving, or deleting objects.

5.1 CloudTrail Trail Configuration

A dedicated CloudTrail trail named **SecurityLab-CloudTrail** was created for the lab environment.

The trail was configured with:

- **Multi-Region trail:** Enabled
- **Management events:** Read and Write events
- **S3 data events:** Enabled
- **Advanced event selector:** Configured specifically for the lab S3 bucket
- **Log file validation:** Enabled

The S3 data-event configuration was restricted to the lab bucket rather than monitoring all S3 resources. This provides a more focused monitoring scope for the security experiment.

5.2 S3 Data Event Selection

A custom advanced event selector named **S3-Lab-Bucket-Events** was configured to capture object-level activity associated with the lab bucket.

The configuration targets S3 object resources and uses the bucket ARN as the resource scope. This allows events such as **PutObject** to be captured for subsequent security monitoring.

Data events [Info](#)
Data events show information about the resource operations performed on or within a resource. [Additional charges apply](#) [↗](#)

Advanced event selectors are enabled
Use the following fields for fine-grained control over the data events captured by your trail. [Switch to basic event selectors](#)

▼ **Data event: S3** [Remove](#)

Resource type
Choose the resource type for which you want to log data events.
S3 ▼

Log selector template
Custom ▼

Selector name - optional
S3-Lab-Bucket-Events
1,000 character limit

Collect events
Log all events, or choose a template to log specific, filtered events to your trail. You can edit templates later.

Advanced event selectors [Info](#)
Log or exclude events based on the values of advanced event selector fields.

Field	Operator	Value
resources.ARN ▼	starts with ▼	arn:aws:s3::security-lab-fawwaz-2026-569486438195-eu-north-1-an/ ✕

[+ Field](#) [+ Condition](#)

► **JSON view**

[Add data event type](#)

Figure 3. CloudTrail configuration showing S3 data-event logging targeted at the security lab bucket.

Security Significance

CloudTrail provides an audit trail of activity performed against the monitored S3 objects. Capturing these data events is important for detecting potentially suspicious actions and

provides the event source required for the CloudWatch-based detection mechanism implemented in the following stages.

6. CloudWatch Log Integration

AWS CloudTrail was integrated with Amazon CloudWatch Logs to provide centralized access to the security events generated by the lab environment.

A dedicated CloudWatch log group named **/aws/cloudtrail/security-lab** was configured to receive CloudTrail events.

This integration allows CloudTrail events to be viewed and analysed within CloudWatch Logs, providing the foundation for the detection mechanism implemented in the next stage.

6.1 CloudTrail Event Verification

To verify that S3 object-level monitoring was working correctly, a test object was uploaded to the monitored S3 bucket.

The resulting CloudTrail event was inspected in CloudWatch Logs. The event recorded an S3 **PutObject** operation, confirming that object-level activity was being captured successfully.

The screenshot displays the AWS CloudWatch Logs console interface. The breadcrumb navigation at the top shows the path: CloudWatch > Log management > /aws/cloudtrail/security-lab > 569486438195_CloudTrail_eu-north-1_2. The left-hand navigation menu includes sections for CloudWatch, Favorites and recents, AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, Logs (with sub-links for Log Management, Log Analytics, and Log Anomalies), Metrics, Network Monitoring, and Setup. The main content area is titled 'Log events' and contains a search bar with the placeholder text 'Filter events - press enter to search'. Below the search bar, a table lists log events with columns for Timestamp, Ingestion time, Message, Event ID, and Log stream name. The first event is expanded, showing a detailed JSON payload. The payload includes metadata such as 'eventTime', 'eventSource', 'eventName' (PUT_OBJECT), and 'awsRegion'. It also contains a 'requestParameters' section with details about the S3 bucket, the object key, and the request headers, including 'X-Amz-Content-Sha256' and 'X-Amz-Expires'. The 'responseElements' section shows the 'x-amz-server-side-encryption' and 'x-amz-version-id' returned by the S3 service. A 'Back to top' button is located in the bottom right corner of the console area.

```
{
  "creationDate": "2026-10-04T09:28:52Z",
  "mfaAuthenticated": "false"
},
{
  "onBehalfOf": {
    "userId": "i45854c8-00b1-7053-8d33-404f3cfae931",
    "identityStoreArn": "arn:aws:identitystore:405378271903:identitystore/3418d478-80c1-702a-64a6-9bb66ee792a6"
  }
},
{
  "eventTime": "2026-10-04T10:36:16Z",
  "eventSource": "s3.amazonaws.com",
  "eventName": "PUT_OBJECT",
  "awsRegion": "eu-north-1",
  "sourceIPAddress": "82.167.42.163",
  "userAgent": "[Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:157.0) Gecko/20100101 Firefox/157.0]",
  "requestParameters": {
    "X-Amz-Date": "20261004T103700Z",
    "bucketName": "security-lab-fawmaz-2026-569486438195-eu-north-1-an",
    "X-Amz-Algorithm": "AWS4-HMAC-SHA256",
    "x-amz-acl": "bucket-owner-full-control",
    "X-Amz-SignedHeaders": "content-type;host;x-amz-acl;x-amz-checksum-crc64nm;x-amz-storage-class",
    "Host": "security-lab-fawmaz-2026-569486438195-eu-north-1-an.s3.eu-north-1.amazonaws.com",
    "X-Amz-Content-Sha256": "UNSIGNED-PAYLOAD",
    "X-Amz-Expires": "300",
    "key": "cloudwatch-test.txt",
    "x-amz-storage-class": "STANDARD"
  },
  "responseElements": {
    "x-amz-server-side-encryption": "AES256",
    "x-amz-version-id": "rcCQ3atCjLs.EE6Rq.0YsRfdBwUeNbfB"
  },
  "additionalEventData": {
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "bytesTransferredIn": 26,
    "AuthenticationMethod": "QueryString",
    "turnAroundTime": 30,
    "bytesTransferredOut": 0,
    "SignatureVersion": "SigV4",
    "Referer": "https://569486438195-k2b56kpf.eu-north-1.console.aws.amazon.com/",
    "objectSize": 26,
    "SSEApplied": "Default SSE S3"
  }
}
```

Figure 4. CloudTrail event recorded in CloudWatch Logs showing an S3 **PutObject** operation against the monitored lab bucket.

7. S3 PutObject Detection Using CloudWatch

A CloudWatch metric filter was created to identify **S3 PutObject events** recorded in the CloudTrail log stream.

The metric filter was configured with the following settings:

- **Filter name:** **S3-PutObject-Detection**
- **Filter pattern:** `{ $.eventName = "PutObject" }`
- **Metric namespace:** **SecurityLab**
- **Metric name:** **S3PutObjectCount**
- **Metric value:** **1**

Whenever a CloudTrail event contains the **PutObject** event name, the metric filter detects the event and publishes a value of **1** to the custom CloudWatch metric. This converts the raw CloudTrail event into a measurable security metric that can be used by a CloudWatch alarm.

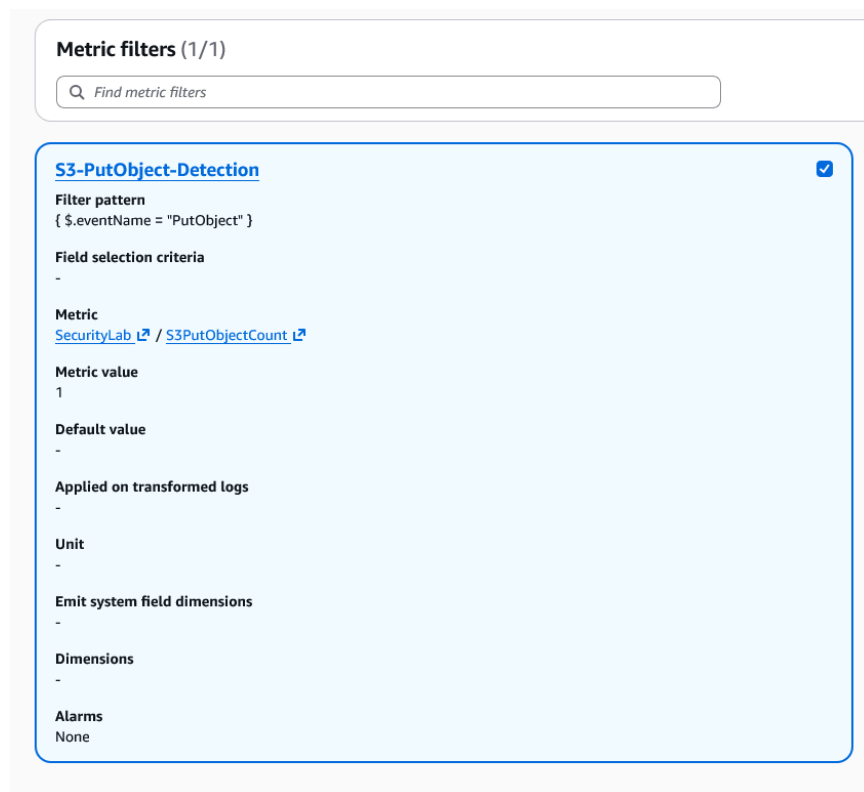


Figure 5. CloudWatch metric filter configured to detect S3 **PutObject** events and publish the **S3PutObjectCount** metric.

8. CloudWatch Security Alarm

A CloudWatch alarm was configured to monitor the custom **S3PutObjectCount** metric generated by the metric filter.

The alarm was configured with the following conditions:

- **Alarm name:** **S3-PutObject-Security-Alert**
- **Metric namespace:** **SecurityLab**
- **Metric:** **S3PutObjectCount**
- **Statistic:** Sum
- **Period:** 5 minutes
- **Threshold:** Greater than or equal to **1**
- **Datapoints to alarm:** 1 out of 1
- **Missing data treatment:** Missing

When at least one **PutObject** event is detected within the configured evaluation period, the metric reaches the defined threshold and the alarm changes to the ALARM state.

The alarm was successfully tested using an S3 object upload, demonstrating that the detection mechanism can identify the configured security event.

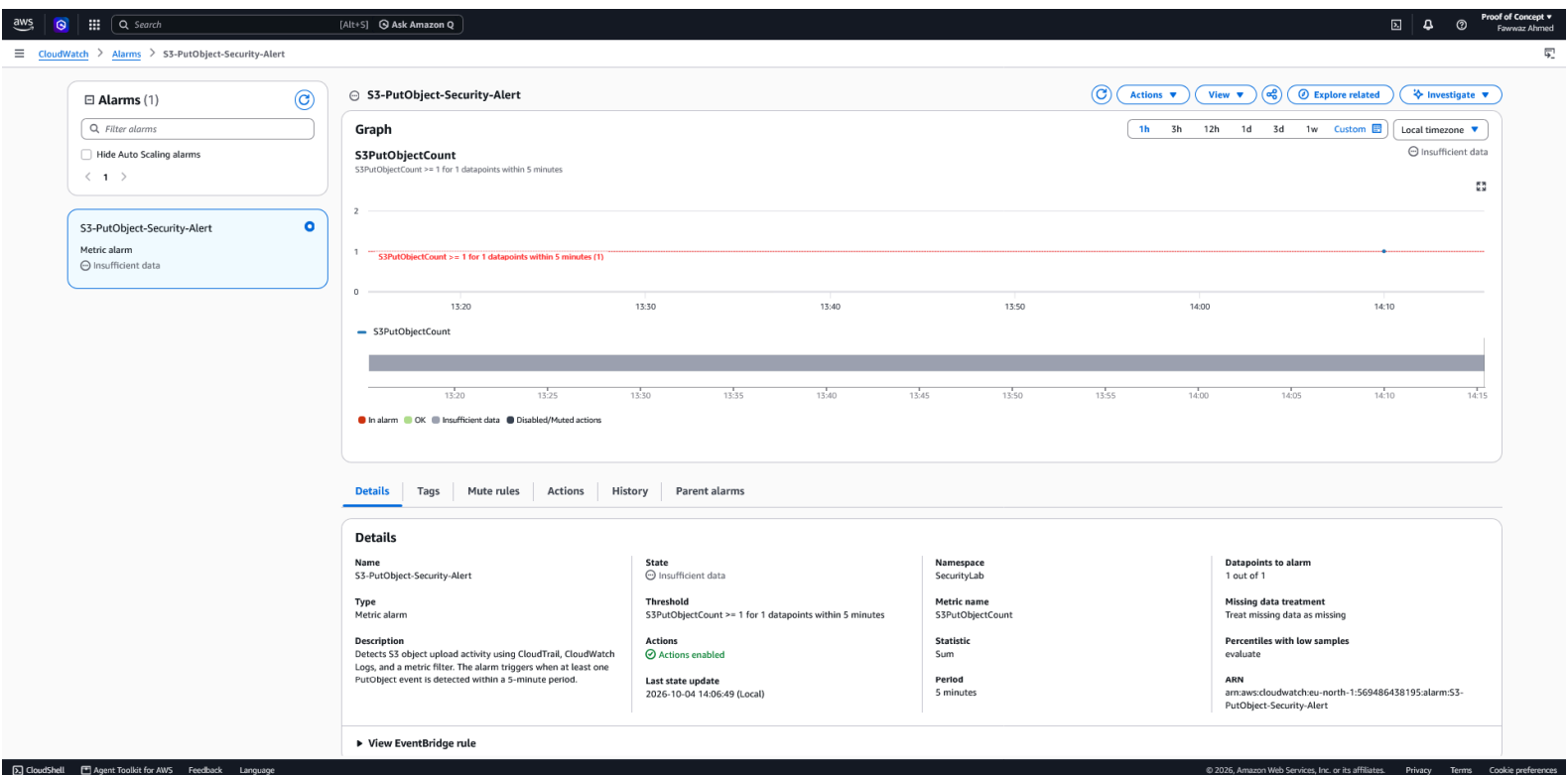


Figure 6. CloudWatch alarm showing the **S3-PutObject-Security-Alert** threshold and the detected **S3PutObjectCount** metric.

9. SNS Email Notification

Amazon SNS was configured to provide an automated notification when the CloudWatch security alarm was triggered.

An SNS topic named **SecurityLab-S3-Alerts** was used as the notification destination for the CloudWatch alarm. An email subscription was configured and confirmed so that security alerts could be delivered directly through email.

When the **S3-PutObject-Security-Alert** alarm entered the ALARM state, SNS automatically generated and delivered an email notification containing details about the alarm and the threshold that had been exceeded.

This demonstrates an automated response mechanism in which a detected cloud security event can generate a notification without requiring continuous manual monitoring of the AWS console.

ALARM: "S3-PutObject-Security-Alert" in EU (Stockholm) Inbox x



AWS Notifications <no-reply@sns.amazonaws.com>
to me ▾

2:15 PM (1 minute ago) ☆ ☺ ↶ ⋮

You are receiving this email because your Amazon CloudWatch Alarm "S3-PutObject-Security-Alert" in the EU (Stockholm) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (04/10/26 11:10:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Sunday 04 October, 2026 11:15:14 UTC".

View this alarm in the AWS Management Console:

<https://eu-north-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=eu-north-1#alarmsV2:alarm/S3-PutObject-Security-Alert>

Alarm Details:

- Name: S3-PutObject-Security-Alert
- Description: Detects S3 object upload activity using CloudTrail, CloudWatch Logs, and a metric filter. The alarm triggers when at least one PutObject event is detected within a 5-minute period.
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (04/10/26 11:10:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Sunday 04 October, 2026 11:15:14 UTC
- AWS Account: 569486438195
- Alarm Arn: arn:aws:cloudwatch:eu-north-1:569486438195:alarm:S3-PutObject-Security-Alert

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: SecurityLab
- MetricName: S3PutObjectCount
- Dimensions:
- Period: 300 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

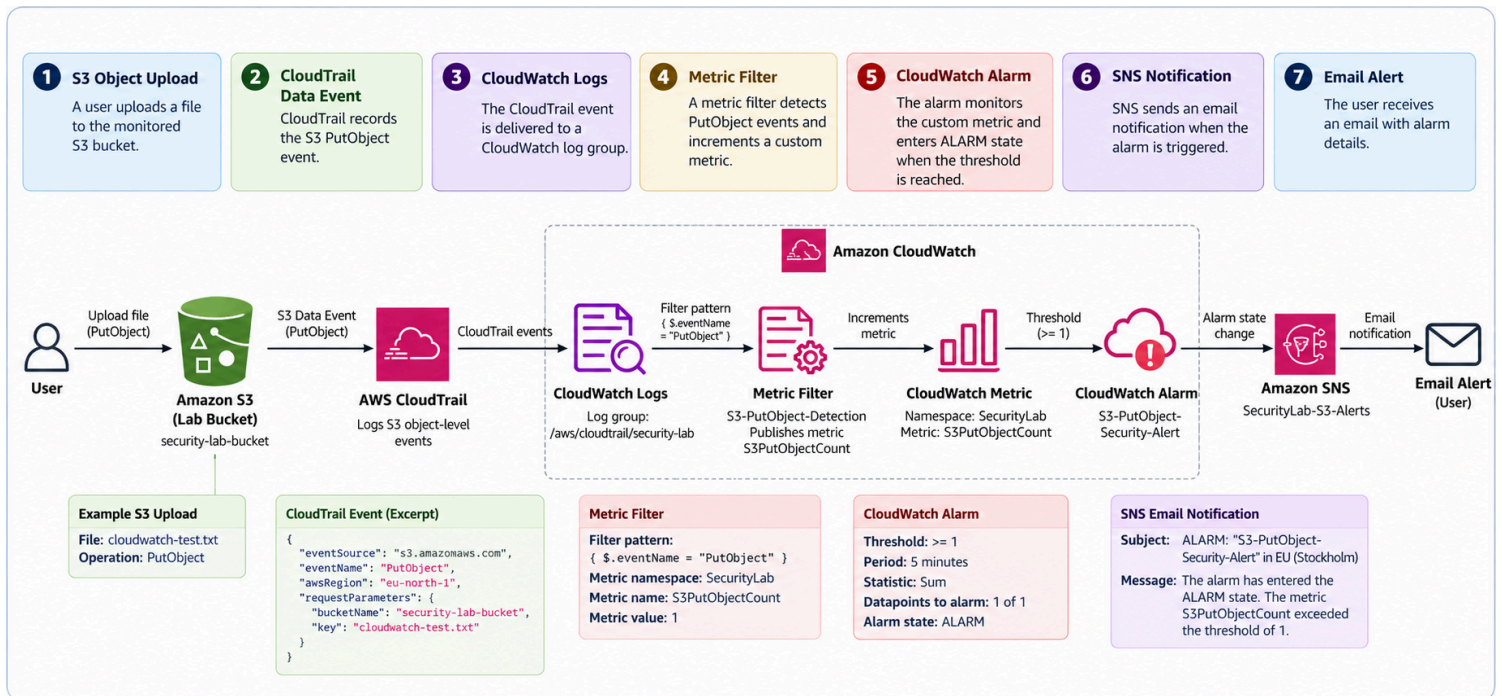
State Change Actions:

- OK:
- ALARM: [arn:aws:sns:eu-north-1:569486438195:SecurityLab-S3-Alerts]
- INSUFFICIENT_DATA:

Figure 7. SNS email notification generated after the CloudWatch alarm detected an S3 **PutObject** event.

10. End-to-End Detection Workflow

The completed AWS security monitoring workflow demonstrates how an S3 object-level activity can be detected and converted into an automated security notification.



11. Security Controls Implemented

The project implemented several security controls across the AWS environment to provide secure storage, activity monitoring, detection, and alerting.

11.1 S3 Security Controls

The S3 bucket was configured with:

- **Block Public Access** enabled to prevent unintended public exposure.
- **Bucket owner enforced** object ownership with ACLs disabled.
- **Versioning** enabled to retain previous object versions.
- **Server-side encryption using SSE-S3.**
- **S3 Bucket Key** enabled.

11.2 Monitoring and Detection Controls

The monitoring architecture included:

- **AWS CloudTrail** for recording S3 object-level data events.
- **CloudWatch Logs** for centralized storage and analysis of CloudTrail events.
- A **CloudWatch metric filter** to identify **PutObject** activity.
- A **CloudWatch alarm** to detect when the configured threshold was reached.

- **Amazon SNS** to provide automated email notification.

Together, these controls provide a layered approach covering data protection, auditing, monitoring, detection, and notification.

12. Cost and Security Considerations

Cost and security were considered throughout the implementation of the lab.

CloudTrail data events were configured specifically for the lab S3 bucket rather than broadly monitoring unnecessary resources. This provides a focused monitoring scope while reducing unnecessary event collection.

An AWS budget was also configured with a low-cost alert threshold to provide notification if AWS spending increased beyond the intended level. The budget acts as a cost-monitoring mechanism rather than a hard spending limit.

The project also avoided exposing unnecessary sensitive information in the public project documentation. Account identifiers, email addresses, credentials, and raw CloudTrail log files should not be published to the public GitHub repository.

The combination of targeted logging, cost monitoring, secure S3 configuration, and controlled documentation helps maintain an appropriate balance between security, visibility, and cost awareness.

13. Skills Demonstrated

This project demonstrates practical skills relevant to cloud security and Security Operations Centre (SOC) roles, including:

- **AWS cloud security**
- **Amazon S3 security configuration**
- **IAM and access-control concepts**
- **CloudTrail audit logging**
- **S3 data-event monitoring**
- **CloudWatch Logs**
- **CloudWatch metric filters**
- **Security alert configuration**
- **Amazon SNS notifications**
- **Security event investigation**
- **Cloud monitoring and detection**
- **Incident-alert workflow design**
- **Security-focused cloud architecture**
- **Cost-aware cloud configuration**

The project also demonstrates the ability to connect individual AWS security services into an end-to-end detection and alerting pipeline, rather than configuring each service independently.

14. Conclusion

This project successfully implemented an AWS cloud security monitoring and alerting environment using Amazon S3, AWS CloudTrail, Amazon CloudWatch, and Amazon SNS.

The lab demonstrated how an S3 `PutObject` activity can be captured as a CloudTrail data event, delivered to CloudWatch Logs, identified using a metric filter, and converted into a CloudWatch alarm. The triggered alarm was then connected to Amazon SNS to generate an automated email notification.

The completed workflow demonstrates a practical approach to cloud security monitoring, event detection, auditing, and automated alerting. It also provided hands-on experience with AWS security controls and technologies relevant to real-world cloud security and SOC environments.